ICT ACADEMY OF KERALA

PROJECT REPORT

STUDENT SKILL AND PROJECT TRACKER

A web-based platform for managing student skills, projects, and progress

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EXECUTIVE SUMMARY

The Student Skill and Project Tracker is a proposed web-based application designed to help students record their skills, organise academic and personal projects, and monitor their learning progress in one place. Students often develop technical and soft skills through coursework, workshops, internships, and team projects, but information about these experiences can become scattered across notebooks, files, and different online platforms. This project aims to provide a centralised and easy-to-use system for maintaining that information.

The application is planned to support student profiles, skill records with proficiency levels, project details, task status, deadlines, and progress summaries. It can help students identify strengths, recognise areas for improvement, and present their experience more clearly to teachers, mentors, or potential employers. The report describes the project's purpose, objectives, scope, methodology, planned activities, expected results, and possible future enhancements. The final implementation and results should be updated to reflect the features actually developed and tested by the project team.

INTRODUCTION

Students take part in a variety of academic activities and practical projects during their studies. These activities help them gain skills such as programming, communication, teamwork, problem-solving, design, research, and time management. However, students may not always maintain a structured record of the skills they have gained or the projects in which they have participated.

The Student Skill and Project Tracker addresses this issue by providing a single platform where students can maintain a personal skills portfolio and track project work. A structured record makes it easier to review progress over time, keep project information organised, and prepare summaries for academic reviews or placement opportunities.

The system is intended to be simple, accessible, and organised around two main areas: skill tracking and project tracking. Skill tracking stores skill names, proficiency levels, and evidence or notes. Project tracking stores project titles, descriptions, technologies, team roles, milestones, deadlines, and current status. Depending on the final implementation, the application may also include dashboards and administrator or mentor functions.

OBJECTIVES

1. Develop a centralised platform for recording and updating students' skills.
2. Allow students to create and maintain project records with descriptions, technologies, roles, and dates.
3. Provide a way to track tasks, milestones, deadlines, and project completion status.
4. Help students review their progress through a clear dashboard or summary view.
5. Encourage students to reflect on skill gaps and plan future learning activities.
6. Apply software development practices such as requirement analysis, interface design, implementation, and testing.

SCOPE AND DELIVERABLES

The project focuses on the design and development of a student-facing application for managing skill and project information. The exact features depend on the implementation completed by the team.

Core scope

- Student profile: basic profile information and an overview of learning activities.
- Skill management: add, edit, view, and remove skills; record a proficiency level and optional evidence or notes.
- Project management: create project entries containing title, description, technologies used, role, start date, target date, and status.
- Task and milestone tracking: maintain a list of tasks and indicate whether each task is pending, in progress, or completed.
- Progress overview: display counts or summaries of skills and projects, along with project status.
- Responsive interface: make the application usable on common desktop and mobile screen sizes.

Expected deliverables

- A working prototype or application of the Student Skill and Project Tracker.
- User interface pages for the main skill and project workflows.
- Data model and storage for student, skill, project, and task information, if supported by the chosen stack.
- Test cases and a brief record of testing outcomes.
- Project documentation describing the design, features, limitations, and future improvements.

METHODOLOGY

A step-by-step development approach is suitable for this project. Requirements are identified first, followed by planning, interface design, implementation, testing, and documentation. The team can use an iterative process, reviewing each feature after implementation and making improvements based on testing or user feedback.

7. Requirement analysis: identify intended users, key problems, and the information that must be recorded.
8. System design: plan the page structure, navigation, data fields, and relationships between students, skills, projects, and tasks.
9. Interface design: prepare simple layouts for the dashboard, skill list, project list, project details, and forms.
10. Implementation: build the interface and connect it to the selected data-storage method or backend service.
11. Testing: check forms, navigation, data validation, editing and deletion, status updates, and responsive behaviour.
12. Review and documentation: record completed features, known limitations, and possible enhancements.

Possible technology stack

The final report should list the tools actually used. One possible implementation is ReactJS for the user interface, Node.js and Express for backend APIs, and MongoDB for storing records. HTML, CSS, and JavaScript support the interface, while Git can be used for version control. If the team uses a different framework or database, replace this suggested stack with the actual tools.

PROJECT ACTIVITIES

1. Requirement gathering and planning

The initial activity is to identify the information students need to maintain. The main records include skills, projects, tasks, milestones, and progress status. The team should define which fields are required, how users navigate between pages, and whether authentication or mentor access is included.

2. Designing the application

The interface can be organised into a dashboard and separate sections for skills and projects. Forms should use clear labels and validation. Lists should present information consistently, while project detail pages can show tasks, dates, technologies, and completion status.

3. Developing skill tracking

The skill module allows a student to enter a skill, select a proficiency level such as Beginner, Intermediate, or Advanced, and optionally add evidence or notes. The interface should allow existing entries to be reviewed and updated so the portfolio remains current.

4. Developing project tracking

The project module records the project name, summary, technologies, role, dates, and status. A task list can divide a project into smaller activities and show whether each activity is pending, in progress, or complete. This structure helps students organise work and monitor deadlines.

5. Dashboard and progress summary

A dashboard can summarise the number of skills recorded, projects created, and tasks completed. These summaries should be based on stored records and should not imply performance ratings that the system does not actually calculate.

6. Testing and refinement

Testing should include valid and invalid form submissions, empty fields, updates to existing records, removal of records, status changes, navigation, and display on different screen sizes. Any defects found should be documented and corrected where possible.

Challenges to consider

- Keeping skill and project information consistent when records are edited.
- Designing a simple interface that can handle several types of information.
- Validating dates, required fields, and task statuses.
- Protecting student information if accounts and online storage are implemented.
- Ensuring the dashboard reflects the latest saved data.

RESULTS & FINDINGS

The intended outcome is a unified application that makes student skills and project activities easier to organise and review. If the planned modules are implemented, students can maintain a skill portfolio, keep project details together, and check the status of tasks from a central dashboard.

The project also demonstrates how a practical information-management problem can be translated into software requirements and user workflows. The usefulness of the application depends on accurate data entry, clear navigation, reliable storage, and appropriate testing.

Suggested testing checklist

- A student can add a skill and see it in the skills list.
- A student can edit or remove an existing skill.
- A student can create a project with the required details.
- Project task statuses can be updated and displayed correctly.

- Required fields and invalid input are handled clearly.
- Pages remain usable on desktop and mobile screens.

Actual test outcomes, screenshots, measured performance, and user feedback should be added after testing. Do not report a feature as successful unless it has been implemented and verified.

CONCLUSION

The Student Skill and Project Tracker is intended to provide students with a structured way to manage their skills and project work. By bringing skill records, project descriptions, tasks, and progress summaries together, the application can support better organisation, reflection, and preparation for future academic or professional opportunities.

The project provides experience in requirement analysis, user-interface design, application development, data management, and software testing. Future development could include secure user authentication, mentor feedback, file or certificate uploads, project collaboration, exportable portfolios, notifications for deadlines, and analytics showing progress over time. These features should be added according to user needs and available development time.

APPENDIX

Appendix A: Suggested data fields

- Student: student ID, name, email, course or department.
- Skill: skill ID, student ID, skill name, proficiency level, evidence or notes, last updated date.
- Project: project ID, student ID, title, description, technologies, role, start date, target date, status.
- Task: task ID, project ID, task title, due date, status, notes.

Appendix B: Suggested screenshots to include

- Home page or dashboard.
- Add/edit skill form and skills list.
- Project creation form and project detail page.
- Task list showing different status values.
- Responsive view on a mobile screen.
- Testing evidence or sample test-case table.

Appendix C: Sample test-case table

Test case	Expected result	Actual result
Add a skill with valid details	Skill is saved and displayed	[Fill after testing]
Submit a form with required fields missing	Validation message is shown	[Fill after testing]
Update a project status	New status is displayed	[Fill after testing]
Open the application on a mobile screen	Content remains readable and usable	[Fill after testing]